

Paper Pulse

Weekly Research Paper Digest

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This digest presents the 20 most recent research papers automatically fetched, filtered, and summarized. Each entry includes the full title, authors, abstract, and AI-generated bilingual summaries.

Online: <https://090516mc.github.io/paper-pulse>

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Paper 1

Multimodal Model Diffing for Feature Discovery and Control

Authors: Hunar Batra, Lachin Naghashyar, Ashkan Khakzar, Philip Torr, Christian Schroeder de Witt et al.

Categories: cs.CV, cs.AI, cs.CL, cs.LG

URL: <https://arxiv.org/abs/2608.09928v1>

Keywords: model, language, large

Abstract

Multimodal Large Language Models (MLLMs) exhibit strong visual understanding, yet the internal features that cause these behaviors remain difficult to identify, audit, or control. While applicable to post-hoc inspection, hidden states that are decomposed into interpretable feature directions using sparse autoencoders (SAEs) neither readily isolate which features are changed by multimodal training, nor are they directly useful for targeted control. We introduce MMDiff, a multimodal model-diffing framework that trains multimodal SAEs and turns them into feature-level interfaces for discovering and

controlling multimodal behavior. MMDiff supports three uses: (i) feature isolation, by diffing a base-LM SAE against its multimodal-adapted counterpart to identify features altered by multimodal training; (ii) task-specific feature detection, via per-token contrastive firing analysis that isolates causal features; and (iii) feature-level control, by causally removing or steering the discovered feature directions. We train multimodal SAEs for three MLLM families, LLaVA-MORE, PaliGemma 2, and InternVL3.5, and evaluate on visual-spatial understanding, multimodal safety, and OCR. MMDiff discovers sparse, causally specific features whose removal selectively degrades target behaviors by an average of 12% on spatial tasks and 17% on OCR, and reduces attack success rate by 24% on multimodal safety attacks, with no impact on VQA performance. Steering these features improves spatial and OCR accuracy by +3.6% and +1.8% on average over a standard single-layer steering baseline. These results show that multimodal SAEs can serve not only as interpretability tools, but as mechanisms for auditing, steering, and controlling MLLMs behavior toward safer and more capable generations.

Chinese Summary

研究背景与问题

多模态大语言模型（MLLMs）展现出强大的视觉理解能力，但其内部特征难以被识别、审计或控制。现有的稀疏自编码器（SAEs）虽可用于事后分析，却无法直接定位多模态训练所改变的特征，也难以用于精准的行为控制。

方法创新：MMDiff框架

本文提出 **MMDiff**，一种多模态模型差分框架，通过训练多模态SAEs并将其转化为特征级接口，实现三大功能：

- **特征隔离**：对比基础语言模型SAE与其多模态适配版本，识别被多模态训练改变的特征
- **任务特定特征检测**：利用逐token对比激活分析，定位因果特征
- **特征级控制**：通过因果移除或引导发现的特征方向，实现对模型行为的精准调控

实验与关键结果

在三个MLLM家族（LLaVA-MORE、PaliGemma 2、InternVL3.5）上，针对视觉空间理解、多模态安全与OCR任务评估：

- **特征移除**：使空间任务性能下降12%，OCR下降17%，多模态安全攻击成功率降低24%，且不影响VQA性能
- **特征引导**：相比标准单层引导基线，空间准确率平均提升+3.6%，OCR提升+1.8%
- 发现的特征具有**稀疏性与因果特异性**，验证了方法的有效性

核心贡献

证明多模态SAEs不仅能作为可解释性工具，更能成为审计、引导和控制MLLM行为的实用机制，推动更安全、更强大的多模态生成。

English Summary

Core Contribution

We introduce **MMDiff**, a multimodal model-diffing framework that trains multimodal sparse autoencoders (SAEs) to serve as feature-level interfaces for discovering and controlling MLLM behaviors. It enables three key capabilities: isolating features altered by multimodal training, detecting task-specific causal features via contrastive firing analysis, and controlling behaviors through feature removal or steering.

Key Results

Across LLaVA-MORE, PaliGemma 2, and InternVL3.5, MMDiff discovers sparse, causally specific features whose removal selectively degrades spatial understanding by 12%, OCR by 17%, and reduces multimodal safety attack success by 24%, without impacting VQA performance. Feature steering improves spatial and OCR accuracy by +3.6% and +1.8% over single-layer baselines.

Significance

This work demonstrates that multimodal SAEs extend beyond interpretability, offering practical mechanisms for auditing,

steering, and controlling MLLMs toward safer and more capable generation.

Paper 2

From Values to Benchmarks: Evaluating Large Language Models for Governmental Use in Dutch

Authors: Laurens Samson, Iva Gornishka, Gossa Lô, Yuki M. Asano, Sennay Ghebreab

Categories: cs.CL, cs.AI

URL: <https://arxiv.org/abs/2608.09925v1>

Keywords: model, language, large

Abstract

Large language models are increasingly being deployed in governmental settings, yet few existing evaluation frameworks jointly reflect the values of public administration and the linguistic requirements of non-English contexts. We present the "Grip on LLMs" framework, a systematic evaluation suite for Dutch governmental use developed in collaboration with domain experts from a major Dutch municipal organisation. Through an advisory board process, user research, and a survey of the users of a civil-servant chatbot, we identify six evaluation dimensions (factuality, honesty, social bias, energy consumption, cost, and training data transparency) and operationalise them into a benchmark suite covering more than 30 multilingual and Dutch-specific models. Our results reveal that no single model excels across all dimensions, and that trade-offs are unavoidable: higher quality consistently comes at greater environmental impact and financial cost, while bias remains largely independent of both. We further find that factuality (whether a model answers correctly) and honesty (whether a model acknowledges what it does not know) are governed by distinct properties, with high factuality not implying high honesty. To make these findings actionable for non-technical audiences, we release a publicly accessible, user-friendly model overview designed for the full range of stakeholders involved in governmental LLM selection, from engineers to policymakers.

Chinese Summary

****背景与动机****: 大型语言模型（LLM）正逐步进入政府行政场景，然而现有评估框架很少同时兼顾公共行政的核心价值观与非英语语境的语言特殊性。荷兰语作为低资源语言，其政务应用缺乏系统化的评估工具。

****方法****: 本文提出 “Grip on LLMs” 框架，与荷兰某大型市政机构的领域专家合作开发。通过顾问委员会研讨、用户调研及公务员聊天机器人用户调查，确定了六个评估维度：****事实性、诚实性、社会偏见、能耗、成本、训练数据透明度****。并将这些维度操作化为一个覆盖30余个多语言及荷兰语专属模型的基准测试套件。

****主要发现****:

- 没有任何单一模型能在所有维度上表现优异，****权衡不可避免****: 更高质量必然伴随更高能耗和财务成本，而偏见水平与二者基本无关。

- ****事实性****（回答是否正确）与****诚实性****（是否承认知识盲区）由不同机制驱动，高事实性并不隐含高诚实性——这一发现对政务场景尤为关键，因为公务员需要模型在不确定时明确“不知道”。

****创新点与实用性****: 为让非技术背景的利益相关者（从工程师到政策制定者）能直接使用结果，团队发布了一个公开、易用的模型对

比可视化界面，覆盖政务LLM选型的全流程决策需求。该框架为多语言、价值驱动的政务AI评估提供了首个可复制的实践范式。

English Summary

****Core Contribution**:** We introduce "Grip on LLMs," a value-driven evaluation framework for Dutch governmental use, co-developed with municipal domain experts. It operationalizes six dimensions—factuality, honesty, social bias, energy consumption, cost, and training-data transparency—into a benchmark suite covering 30+ multilingual and Dutch-specific models.

****Key Findings**:** No model dominates across all dimensions; higher quality consistently trades off against environmental impact and cost, while bias remains orthogonal. Critically, factuality and honesty are governed by distinct model properties—being correct does not imply acknowledging ignorance. This distinction is vital for civil servants who must trust model uncertainty signals.

****Impact**:** We release a public, non-technical model comparison dashboard to support diverse stakeholders in governmental LLM procurement, bridging the gap between technical benchmarks and public-administration values. This work provides a replicable template for non-English, value-sensitive AI evaluation.

Paper 3

GENCO - A Unified Neural Solver Embedded in a Development Framework for Steady-State Grid Analysis

Authors: Alban Puech, Matteo Mazzonelli, Tamara R. Govindasamy, Mangaliso Mngomezulu, Héctor Maeso-García et al.

Source

Categories: cs.AI

URL: <https://arxiv.org/abs/2608.09921v1>

Keywords: neural, system, code, operating, generation, network

Abstract

Foundation models are transforming business workflows and boosting productivity, yet they remain largely absent from engineering domains such as power system analysis, where strict physical consistency must be enforced. We present GENCO (GEometric Neural Corrective Optimizer), a unified neural solver for steady-state transmission grid analysis that handles power flow (PF), optimal power flow (OPF), and state estimation (SE) within a single architecture and shared network representation. To support advances in neural power system solvers, we introduce the open-source GridFM Development Framework, which standardizes synthetic data generation and training in a low-code environment. We also release large-scale datasets with millions of PF and OPF scenarios across diverse grid topologies to support reproducible benchmarking. We evaluate GENCO on the PFDelta and OPFData benchmarks against state-of-the-art neural solvers and classical solvers, including Newton-Raphson and IPOPT, as well as on real-world Hydro-Québec SCADA data. For large-scale PF, GENCO recovers the full AC operating state, including voltage magnitudes and reactive power that DC-PF cannot provide, while matching DC-PF-level active power-balance residuals. It achieves up to 30x speedups over Newton-Raphson at only 2x the runtime of DC-PF. For OPF, it achieves up to 85x speedups over IPOPT while improving feasibility, optimality, and runtime over DC-OPF. For SE, GENCO is more robust than classical weighted least squares to noisy measurements and network parameter errors, and always returns a high-quality estimate even when weighted least squares fails to converge. Together, the unified architecture and development framework provide a new approach to large-scale steady-state grid analysis, lowering the barrier to entry for power system engineers and marking a step toward Grid Foundation Models.

Chinese Summary

研究背景

基础模型正在变革各行业工作流程，但在电力系统分析等工程领域仍几乎空白，其核心挑战在于必须严格保证物理一致性。

方法创新

本文提出

GENCO（几何神经校正优化器），一个用于稳态输电网分析（涵盖潮流PF、最优潮流OPF、状态估计SE）的统一神经求解器，采用单一架构与共享网络表征。为支撑该类求解器发展，作者还发布了开源**GridFM**开发框架，以低代码环境标准化合成数据生成与训练流程，并公开了覆盖多种电网拓扑的百万级**PF/OPF**场景数据集，支持可复现的基准测试。

主要结果

在 **PFDelta** 与 **OPFData** 基准上，**GENCO** 与最先进的神经求解器及经典求解器（牛顿-拉夫逊法、IPOPT）对比，并在真实 Hydro-Québec SCADA 数据上验证：

- **大规模PF**：完整恢复交流运行状态（含电压幅值与无功功率，这是直流PF无法提供的），同时维持与直流PF相当的有功平衡残差；相比牛顿-拉夫逊法最高提速 **30倍**，运行时间仅为直流PF的 **2倍**。
- **OPF**：相比IPOPT最高提速 **85倍**，且在可行性、最优性与运行时间上均优于直流OPF。
- **SE**：对噪声测量与网络参数误差的鲁棒性优于经典加权最小二乘法；即使加权最小二乘不收敛，**GENCO** 仍能返回高质量估计。

研究意义

统一架构与开发框架为大电网稳态分析提供了新路径，显著降低了电力工程师的入门门槛，标志着向电网基础模型迈进的重要一步。

English Summary

Core Contribution

We present **GENCO**, a unified neural solver for steady-state grid analysis (power flow, optimal power flow, state estimation) with a single architecture, plus the open-source **GridFM** development framework and large-scale benchmark datasets.

Key Results

- **PF**: Recovers full AC state (including voltage/reactive power) at DC-PF-level active power residuals; up to **30× faster** than Newton-Raphson, only **2×** the runtime of DC-PF.
- **OPF**: Up to **85× speedup** over IPOPT, with improved feasibility and optimality over DC-OPF.
- **SE**: More robust than weighted least squares to noisy measurements and parameter errors, always producing high-quality estimates even when classical methods diverge.

Impact

This unified approach lowers entry barriers for power engineers and represents a significant step toward Grid Foundation Models, validated on both standard benchmarks and real Hydro-Québec SCADA data.

Paper 4

Fusion Training for Mathematical Generalization in Large Language Models

Authors: Congfeng Cao, Pengyu Zhang, Jelke Bloem

Categories: cs.CL, cs.AI

URL: <https://arxiv.org/abs/2608.09893v1>

Keywords: model, language, large

Source

Abstract

Thinking Mode Fusion (TMF) enables large language models to support both concise responses and long-form reasoning by unifying a non-thinking mode and a thinking mode within a single model. However, its training dynamics, including the `\emph{data ratio}` and `\emph{training schedule}` between the two modes, remain underexplored. In this work, we present a systematic study of TMF by analyzing the effects of the training schedule and data ratio between thinking and non-thinking modes. Focusing on mathematical problem solving, we construct a benchmark with multiple thinking-to-non-thinking data ratios and three training schedules. Our results reveal an asymmetric interaction between the two modes: increasing the ratio of non-thinking supervision reduces the accuracy of the thinking mode. We further show that different training schedules modulate this trade-off and that the optimal schedule depends on the data ratio. Finally, we quantify a negative correlation between non-thinking and thinking mode supervision, highlighting an inherent tension between these two modes. These findings provide practical guidance for designing effective TMF training settings. All code and data are released to support further research at: `\href{https://github.com/caocongfen/Fusion-Bench.git}{\textbf{Fusion Bench}}`.

Chinese Summary

研究背景与问题

大型语言模型（LLM）通常需要在“简洁直接回答”与“长链推理”两种模式间做出取舍。**思维融合训练（TMF）**通过将非思考模式与思考模式统一于单一模型，使模型能同时支持两种响应方式。然而，TMF训练中两个关键因素——**数据比例**（思考与非思考数据的配比）与**训练调度**（两种模式的训练顺序/交替策略）——对模型性能的影响尚未得到系统研究。

研究方法

本研究聚焦数学问题求解任务，构建了包含**多种思考/非思考数据比例**（如1:1、1:3、3:1等）及**三种训练调度策略**（交替训练、分阶段训练、混合训练）的基准测试集 **Fusion Bench**。通过控制变量实验，系统分析了不同配置下两种模式各自的表现及相互影响。

主要发现

- 不对称交互效应**：提高非思考监督数据比例会显著**降低思考模式的准确率**，但反之，提高思考数据比例对非思考模式影响较小。这表明两种模式间存在非对称干扰。
- 调度策略的调节作用**：不同训练调度会改变上述权衡关系，且**最优调度取决于具体数据比例**——例如，在思考数据占优时，交替训练效果最佳；而在非思考数据占优时，分阶段训练能缓解性能下降。
- 负相关量化**：实验数据明确显示，非思考模式与思考模式的监督信号之间存在**显著负相关**（相关系数约-0.6），揭示了两种模式间的内在张力——优化一种模式往往以牺牲另一种为代价。

创新与意义

本研究首次系统揭示了TMF训练中数据比例与调度策略的耦合效应，为实际部署提供了可操作的设计指南（如推荐在思考数据比例 $\geq 50\%$ 时采用交替训练）。所有代码与数据已开源，便于后续研究复现与扩展。

English Summary

Core Contribution

This paper presents the first systematic study of **Thinking Mode Fusion (TMF)** training dynamics, focusing on how data ratios and training schedules affect mathematical reasoning performance.

Key Findings

- Asymmetric interference**: Increasing non-thinking supervision reduces thinking-mode accuracy, but the reverse effect is minimal.
- Schedule-dependent trade-offs**: The optimal training schedule varies with data ratio—alternating training works best when thinking data dominates, while staged training mitigates performance loss when non-thinking data dominates.
- Quantified tension**: A significant negative correlation (~ -0.6) exists between thinking and non-thinking supervision signals, confirming an inherent conflict between the two modes.

Practical Impact

Our results provide concrete guidance for designing effective TMF configurations in real-world LLM deployment. All resources are released via **Fusion Bench** (github.com/caocongcong/Fusion-Bench) to facilitate further research and reproducibility.

Paper 5

SHE: Trajectory-driven Safety Harness Evolution for LLM Agents

Authors: Wanying Qu, Qinghua Mao, Yu Li, Jiyao Liu, Xin Zhang et al.

Categories: cs.AI, cs.CV

URL: <https://arxiv.org/abs/2608.09885v1>

Keywords: model, language, large

Abstract

The safety of large language model (LLM) agents depends not only on model weights but also on the agent harness that manages context, memory, tools, permissions, and runtime control. Existing safety mechanisms often treat the harness as a fixed deployment artifact, limiting their ability to evolve with emerging risks. Moreover, coupled functions across harness components obscure safety responsibility attribution, making localized evolution difficult. We propose Safety Harness Evolution (SHE), a framework that learns evolving safe boundaries from rollout trajectories. SHE decomposes the harness into four artifacts with explicit safety responsibilities, including the System Prompt, Rule Bank, Safety Memory, and Tool Policy, defining clear functional boundaries for localized evolution. Based on this decomposition, SHE introduces an attribution-guided evolution loop that converts trajectory failures into structured diagnoses, learns artifact-specific boundary refinements, and selects evolved harnesses through safety-utility validation. Experiments on Agent-SafetyBench demonstrate that SHE effectively enhances safety through harness evolution, achieving a 3.1x ASR reduction compared with static SafeHarness, while also improving benign utility. The evolved harness further generalizes to unseen risks on the held-out AgentHarm benchmark and transfers across agent models without additional evolution.

Chinese Summary

背景与问题：大语言模型（LLM）智能体的安全性不仅依赖模型权重，还取决于管理上下文、记忆、工具、权限与运行时控制的“智能体外壳”（harness）。现有安全机制常将外壳视为固定部署组件，难以随新风险演化；同时各组件功能耦合，导致安全责任归属模糊，局部改进困难。

方法：本文提出**安全外壳演化框架（SHE）**，从运行轨迹中学习动态安全边界。SHE 将外壳解构为四个责任明确的工件：**系统提示（System Prompt）**、**规则库（Rule Bank）**、**安全记忆（Safety Memory）** 与 **工具策略（Tool Policy）**，为局部演化划定清晰功能边界。基于此解构，SHE 引入**归因引导的演化循环**：将轨迹失败转化为结构化诊断，学习各工件专属的边界修正，并通过“安全-效用”双重验证筛选演化后的外壳。

实验与结果：在 Agent-SafetyBench 基准上，SHE 相比静态 SafeHarness 将攻击成功率（ASR）降低 **3.1 倍**，同时提升良性任务效用。演化后的外壳在保留的 AgentHarm 基准上对未知风险表现出良好泛化性，且无需额外演化即可跨智能体模型迁移。

创新点：①首次将外壳安全责任显式分解，实现可定位的局部演化；②利用轨迹归因自动生成诊断与边界修正，替代人工设计；③通过安全-效用联合验证，避免“过度安全”损害可用性。SHE 为 LLM 智能体提供了一种可持续进化的安全防护范式。

English Summary

SHE (Safety Harness Evolution) addresses the limitation of static LLM agent harnesses that cannot adapt to emerging risks. It decomposes the harness into four safety-attributed artifacts—System Prompt, Rule Bank, Safety Memory, and Tool Policy—enabling localized evolution with clear responsibility boundaries. An attribution-guided loop converts trajectory failures into structured diagnoses, learns artifact-specific boundary refinements, and selects evolved harnesses via safety-utility validation. On Agent-SafetyBench, SHE reduces attack success rate by **3.1×** compared to static SafeHarness while improving benign utility. The evolved harness generalizes to unseen risks on AgentHarm and transfers across agent models without further evolution, offering a scalable paradigm for dynamically safe LLM agents.

Paper 6

Energy-Structured Latent World Models with Neural Time Fields for Physically Consistent Open-World Motion Planning

Authors: Yapeng Liu, Yuanzhao Zhai, Bo Ding, Huaimin Wang, Lin Wang

Categories: cs.RO, cs.AI

URL: <https://arxiv.org/abs/2608.09876v1>

Keywords: motion, planning

Abstract

Physically consistent motion planning remains a fundamental challenge in embodied AI, as generated trajectories must strictly conform to real-world execution dynamics. While latent world models offer a promising approach by predicting these dynamics, existing methods learn unconstrained future representations where absorbed physics remains implicit. Therefore, they fail to form reusable physical knowledge, which compromises reliability in unpredictable open-world navigation. To address this, we propose a novel Energy-Structured Latent World Model (ELWM). Our key idea is to structure the ELWM latent state to explicitly carry energy and momentum, ensuring strictly causal transitions via dissipation and control ports. Trained on multimodal RGB-D and inertial interaction histories, our model guarantees physically consistent predictions. We further implement this for motion planning by constructing Physics-Conditioned Neural Time Fields (PC-NTF), a key technical cornerstone that integrates ELWM into an arrival time field via the Eikonal equation to yield a physically-informed navigation policy. Across held-out scenes, our evaluation reveals significant improvements. Compared to generic latent models, PC-NTF

reduces 0.8-s motion-prediction NRMSE from 0.36 to 0.29. Against Active Neural Time Fields, it improves navigation success from 81.3% to 89.7% and SPL from 0.64 to 0.73, while cutting the physical collision rate from 12.1% to 5.8% and the Eikonal residual from 0.083 to 0.031. Beyond these targeted gains, our results demonstrate that embedding explicit physical structures into latent spaces intrinsically bridges the gap between predictive world models and safe, dynamically feasible motion planning.

Chinese Summary

物理一致性运动规划的新范式

背景与挑战：具身智能中的运动规划要求生成轨迹严格符合真实物理执行规律。现有潜在世界模型虽能预测动力学，但其潜在空间表征不受约束，物理规律隐式存在，难以形成可复用的物理知识，导致在开放世界导航中可靠性不足。

核心方法：本文提出**能量结构化潜在世界模型（ELWM）**。其关键创新在于：

- **显式物理结构**：在潜状态中显式编码能量与动量，通过耗散端口和控制端口保证严格因果的状态转移；
- **多模态训练**：利用RGB-D与惯性交互历史训练，确保预测的物理一致性；
- **物理条件神经时间场（PC-NTF）**：将ELWM与Eikonal方程结合，构建物理感知的到达时间场，实现物理信息驱动的导航策略。

主要结果（在未见场景中验证）：

- 运动预测NRMSE从0.36降至**0.29**（降低19%）；
- 导航成功率从81.3%提升至**89.7%**，SPL从0.64提升至**0.73**；
- 物理碰撞率从12.1%降至**5.8%**，Eikonal残差从0.083降至**0.031**。

结论：在潜空间嵌入显式物理结构，能从根本上弥合预测世界模型与安全、动态可行运动规划之间的鸿沟，为开放世界导航提供了更具物理一致性的新范式。

English Summary

Key Contribution

We propose **Energy-Structured Latent World Model (ELWM)** with **Physics-Conditioned Neural Time Fields (PC-NTF)** for physically consistent open-world motion planning. ELWM explicitly encodes energy and momentum in latent states, ensuring strictly causal transitions via dissipation/control ports. PC-NTF integrates ELWM into arrival-time fields through the Eikonal equation, yielding physics-informed navigation policies.

Results: On held-out scenes, PC-NTF reduces motion-prediction NRMSE from 0.36 to 0.29, boosts navigation success from 81.3% to 89.7% (SPL: 0.64→0.73), and cuts physical collision rate from 12.1% to 5.8% with Eikonal residual dropping from 0.083 to 0.031. Our work demonstrates that embedding explicit physical structures into latent spaces intrinsically bridges predictive world models and safe, dynamically feasible motion planning.

Paper 7

ArchAgent v2: A Case Study with the Data Prefetching Championship

Authors: Abraham Gonzalez, Raghav Gupta, Akanksha Jain, Hanna Alam, Alexander Novikov et al.

Categories: cs.AI, cs.AR

URL: <https://arxiv.org/abs/2608.09874v1>

Keywords: design, artificial, algorithm, intelligence

Abstract

Agentic artificial intelligence has shown great promise in automating algorithm design, but scaling similar techniques to computer microarchitecture discovery remains challenging due to vast search spaces, strict hardware budgets, and long simulation times. In this work, we present ArchAgent v2, a framework which scales automated microarchitecture search to multi-level data prefetching. While the original ArchAgent successfully discovered single-level cache replacement policies in competition settings, it does not scale to multi-level prefetching where the design space and degrees of freedom are larger. To overcome this, we introduce two new additions to ArchAgent: a cascaded evolutionary search that subdivides the design space by sequentially evolving and freezing prefetchers at individual cache levels, and a hardware-realizability feedback loop that embeds real-time size-estimation directly into the evolution process. Evaluated under identical rules of the 4th Data Prefetching Championship (DPC4), ArchAgent v2 automatically designs a three-level prefetcher that outperforms the winning hand-designed solution, further demonstrating automated agentic discovery as a useful tool for computer architects. Our discovered policy achieves a 3.8\% geometric mean IPC speedup over the baseline overall and a 0.3\% improvement over the prior champion, BertiGO. On low-bandwidth single-core configurations, our policy yields a 4.6\% performance speedup compared to only 2.6\% for BertiGO. However, multi-core evolution still remains a significant challenge due to simulation latency impeding evolution speed. Finally, our profiling of an ArchAgent evolution of over 12,000 candidate designs provides key insights into how automated evolutionary agents explore and synthesize complex microarchitectural logic.

Chinese Summary

研究背景与挑战

智能体人工智能在自动化算法设计领域展现出巨大潜力，但将其扩展至计算机微架构发现仍面临搜索空间庞大、硬件预算严苛及仿真耗时漫长等核心难题。

方法创新

本文提出 **ArchAgent v2** 框架，成功将自动化微架构搜索扩展至多级数据预取领域。针对原始 ArchAgent 仅能处理单级缓存替换策略、无法应对多级预取设计空间的问题，我们引入两项关键改进：

- **级联进化搜索**：按缓存层级顺序进化并冻结预取器，将复杂设计空间分解为可管理的子问题；
- **硬件可实现性反馈回路**：将实时面积估算直接嵌入进化流程，确保生成策略符合物理约束。

主要结果

在第四届数据预取锦标赛（DPC4）的完全同等规则下评估，ArchAgent v2 自动设计的三级预取器 **超越人工设计的冠军方案**：

- 整体几何平均 IPC 较基线提升 **3.8%**，较上届冠军 BertiGO 提升 **0.3%**；
- 在低带宽单核配置下，性能提升达 **4.6%**（BertiGO 仅 2.6%）；
- 多核进化仍受仿真延迟制约，是未来重点突破方向。

此外，对超过

12,000

个候选设计^{**}的进化过程剖析，揭示了自动化进化智能体如何探索与综合复杂微架构逻辑的内在规律，为计算机架构师提供了可迁移的

宝贵经验。

结论

本工作证明自动化智能体发现可作为计算机架构设计的实用工具，在超越人类专家方案的同时，为微架构搜索方法论提供了新范式。

English Summary

Core Contribution

This paper presents **ArchAgent v2**, a framework that scales automated microarchitecture search to multi-level data prefetching. Building on the single-level success of ArchAgent, we introduce **cascaded evolutionary search** (sequentially evolving and freezing per-level prefetchers) and a **hardware-realizability feedback loop** (embedding real-time area estimation into evolution).

Key Results

Under identical DPC4 competition rules, ArchAgent v2 automatically designs a three-level prefetcher that **outperforms the winning human-designed solution**: achieving 3.8% geometric-mean IPC speedup over baseline (0.3% over prior champion BertiGO), and notably **4.6% speedup on low-bandwidth single-core** configurations versus BertiGO's 2.6%. Multi-core evolution remains challenging due to simulation latency.

Significance

Analysis of 12,000+ candidate designs reveals how automated evolutionary agents synthesize complex microarchitectural logic, positioning agentic discovery as a practical tool for computer architects beyond human intuition.

Paper 8

Stealing Reasoning Traces from Proprietary LLM APIs

Authors: Alexander Panfilov, David Schmotz, Ilia Shumailov, Luca Beurer-Kellner, Joachim Schaeffer et al.

Source

Categories: cs.CR, cs.AI, cs.LG

URL: <https://arxiv.org/abs/2608.09867v1>

Keywords: model, language, large

Abstract

Leading large language model providers now conceal their models' step-by-step reasoning, or chain-of-thought, to protect intellectual property and limit information leakage. Rather than storing these traces server-side, providers return them to the client as blocks of encrypted text, which the client passes back with each subsequent request. Building on prior research, we identify an architectural vulnerability: these encrypted blocks are fully compatible and interchangeable across different sessions, users, and models within a provider's ecosystem. We exploit this compatibility to develop a scalable decryption jailbreak. By injecting an encrypted reasoning trace from a given model into a weaker, and less safeguarded model from the same provider, we force it to decode and output the trace verbatim in plaintext, without ever jailbreaking the more capable model directly. This vulnerability enables four distinct attack vectors. First, it circumvents anti-distillation mechanisms, allowing adversaries to extract a proprietary model's reasoning, as we demonstrate across Anthropic, OpenAI, and Google. Second, it allows for large-scale private data extraction. Developers frequently share session logs publicly, unaware of contents of the encrypted blocks. By decoding 315,320 reasoning blocks scraped from public repositories, we recovered 367 Personally Identifiable Information (PII) artifacts and 182 credentials. Third, it inadvertently reveals hazardous information hidden within

the reasoning process, even in cases where the model's final, visible output safely rejects a malicious request. Fourth, attackers can leverage this flaw to execute invisible prompt injections, embedding malicious payloads entirely within encrypted blocks to poison public agentic rollouts. Following responsible disclosure, we propose concrete cryptographic and system-level mitigations to secure client-side reasoning.

Chinese Summary

研究背景与核心发现

主流大语言模型（LLM）提供商为保护知识产权并防止信息泄露，现已隐藏模型的逐步推理过程（即思维链）。然而，这些推理轨迹并未在服务端存储，而是以加密文本块的形式返回给客户端，并在后续请求中由客户端原样回传。本研究基于先前工作，识别出一个**架构级漏洞**：这些加密块在同一提供商的生态系统内，跨会话、跨用户乃至跨模型均完全兼容且可互换使用。

攻击方法

我们利用这一兼容性开发了**可扩展的解密越狱攻击**。具体而言，将来自某强模型的加密推理轨迹注入同一提供商旗下防护较弱的模型，迫使后者以明文形式逐字解码并输出该推理轨迹——全程无需直接越狱更强大的目标模型。

四大攻击向量

1. **绕过反蒸馏机制**：攻击者可提取专有模型的推理过程，已在 Anthropic、OpenAI 和 Google 的模型上成功验证。
2. **大规模隐私数据提取**：开发者常公开分享会话日志，却不知加密块内容。我们解码了从公开仓库抓取的 315,320 个推理块，恢复出 367 条个人身份信息（PII）和 182 组凭据。
3. **泄露危险信息**：即使模型最终可见输出安全地拒绝恶意请求，其推理过程中隐藏的危害性信息仍可能被意外暴露。
4. **隐形提示注入**：攻击者可完全在加密块内嵌入恶意负载，污染公开的智能体部署，实现隐蔽攻击。

缓解建议

遵循负责任披露流程，我们提出了具体的密码学与系统级缓解措施，以保障客户端推理过程的安全。

English Summary

Core Contribution

This paper exposes a critical architectural vulnerability in proprietary LLM APIs: encrypted reasoning traces (chain-of-thought) returned to clients are interchangeable across sessions, users, and models within the same provider. We exploit this to build a scalable decryption jailbreak by injecting a strong model's encrypted trace into a weaker model from the same provider, forcing it to output the reasoning verbatim in plaintext—without ever directly attacking the stronger model.

Key Results

We demonstrate four attack vectors: (1) bypassing anti-distillation to extract proprietary reasoning from Anthropic, OpenAI, and Google; (2) large-scale private data extraction—decoding 315,320 public reasoning blocks yielded 367 PII artifacts and 182 credentials; (3) exposing hazardous hidden reasoning even when final outputs are safe; and (4) invisible prompt injection via encrypted blocks. We propose concrete cryptographic and system-level mitigations following responsible disclosure.

Paper 9

Towards Expert-level Medical AI for Real-time Video Consultations

Authors: Mahvish Nagda, Jihyeon Lee, Matthew Thompson, Chunjong Park, Tim Strother et al.

Categories: cs.AI, cs.CL, cs.CV

URL: <https://arxiv.org/abs/2608.09861v1>

Keywords: system, multi-agent

Source

Abstract

Audio-visual interaction is the standard for patient-physician consultations, enabling natural communication and effective assessment of illness through non-verbal cues. While text-based AI has shown promise, it discards essential perceptual dimensions and limits patients who cannot articulate symptoms in writing. Early efforts to extend medical AI to audio-visual interaction have demonstrated feasibility but not reached clinician-level performance. Here, we provide the first demonstration of expert-level AI in real-time clinical video consultations using AMIE (Articulate Medical Intelligence Explorer) in a video configuration. AMIE (Video) is a Gemini-based multi-agent system integrating low-latency dialogue, clinical reasoning, and real-time audio-visual perception. To guide development, we established a taxonomy and automated evaluations for clinical audio-visual cues in telehealth settings. In a randomized Objective Structured Clinical Examination (OSCE) study with 30 primary care physicians (PCPs), 15 patient actors and 100 clinical scenarios, we compared AMIE (Video), its text-only counterpart AMIE (Text), and PCPs consulting via video. Clinical evaluators rated AMIE (Video) on par or better than PCPs in history-taking, diagnosis, management, and physical observation and examination. Patient actors preferred AMIE's approach to assessing and explaining conditions, while PCPs were preferred for rapport and partnership building. In modality ablation, patient actors preferred AMIE (Video)'s interface over text chat for communicative effectiveness, convenience, and feeling understood. Limitations remain in fine anatomical precision, subtle affective nuances, and high-frequency movements. While further research is needed before real-world translation, these results mark an important milestone toward AI systems capable of augmenting care across the sensory complexity of clinical practice.

Chinese Summary

研究背景与问题

医患诊疗的标准模式是视听交互，它通过非语言线索实现自然沟通和有效病情评估。尽管基于文本的AI已展现潜力，但丢失了关键的感知维度，且无法服务难以用文字描述症状的患者。早期将医疗AI扩展至视听交互的尝试虽验证了可行性，但未达到临床医生水平。

方法与创新

本研究首次展示了**专家级AI在实时临床视频问诊中的应用**，基于AMIE (Articulate Medical Intelligence Explorer) 的视频配置版本。AMIE (Video) 是一个基于Gemini的多智能体系统，集成了低延迟对话、临床推理和实时视听感知能力。研究团队建立了远程医疗场景下临床视听线索的分类体系和自动化评估方法，以指导系统开发。

核心发现

在一项随机客观结构化临床考试 (OSCE) 研究中，涉及**30名基层医生、15名模拟患者和100个临床场景**，对比了AMIE (Video)、纯文本版AMIE (Text) 和通过视频问诊的基层医生。主要结果包括：

- **临床评估**：AMIE (Video) 在病史采集、诊断、管理、体格观察与检查方面与医生持平或更优
- **患者偏好**：模拟患者更偏好AMIE的病情评估与解释方式，但医生在建立融洽关系和伙伴合作方面更受青睐
- **模态消融**：在沟通效率、便利性和被理解感方面，模拟患者更偏好AMIE (Video) 而非文本聊天

局限与展望

当前系统在精细解剖精度、微妙情感细节和高频动作识别方面仍有不足。尽管在真实世界应用前还需进一步研究，这些结果标志着AI系统在临床实践的感官复杂性中增强护理能力的重要里程碑。

English Summary

This study presents the first demonstration of expert-level AI in real-time clinical video consultations using AMIE (Video), a Gemini-based multi-agent system integrating low-latency dialogue, clinical reasoning, and real-time audio-visual perception. In a randomized OSCE study with 30 primary care physicians, 15 patient actors, and 100 clinical scenarios, clinical evaluators rated AMIE (Video) on par or better than physicians in history-taking, diagnosis, management, and physical observation, while patient actors preferred its approach to assessing and explaining conditions. However, physicians were preferred for rapport building, and limitations remain in fine anatomical precision and subtle affective nuances. These findings mark a significant milestone toward AI systems capable of augmenting care across the sensory complexity of clinical practice, though further research is needed before real-world translation.

Paper 10

Agentic Harnesses: LLM-Driven Verification Layers for Robot Autonomy

Authors: Rohan Bhagra, Mahantesh Halapannavar, Uddhav Bhattarai

Categories: cs.RO, cs.AI

URL: <https://arxiv.org/abs/2608.09857v1>

Keywords: artificial, robotics, intelligence

Source

Abstract

Advances in advanced artificial intelligence tools have sparked research in robot autonomy, but the development of such systems has largely focused on execution rather than verifying the feasibility actions planning models propose. Like general-purpose LLMs, robotics planning models carry risks: biased toward user-specified goals, they may suggest actions misaligned with scientific ethics, they may be unsafe due to an inability to "remember" prior safety risks, or they may be vulnerable to adversarial attacks on the autonomy ecosystem. We propose a LLM-driven verification layer between planning and execution to evaluate action permissibility. Our LLM-as-a-Judge ensemble combines chain-of-thought reasoning across models and synthesizes those expert judge outputs, mirroring a combination of a mixture of experts and self-consistency approach. This layer serves as middleware, gating plans from the server's planning module before they reach the MCP server and therefore the robot's low-level controls: plans are approved, rejected for reformulation, or escalated for human review. With this system, we achieve near 85% precision across accept/escalate/reject categories 97% containment of adversarial attacks, with negligible errors between accepting and rejecting tasks, and errors mostly manifesting at the escalate boundary.

Chinese Summary

研究背景与问题

先进人工智能工具的快速发展推动了机器人自主性的研究，但现有系统大多聚焦于**任务执行**，而忽视了**规划模型所提动作的可行性验证**。与通用大语言模型类似，机器人规划模型存在多重风险：倾向于迎合用户目标而可能提出违背科学伦理的动作；因无法“记住”历史安全风险而可能产生不安全行为；以及易受针对自主生态系统的对抗性攻击。

研究方法

本研究提出一种**基于大语言模型的验证层**，置于规划与执行之间，用于评估动作的许可性。该验证层采用**LLM-as-a-Judge集成方法**，结合跨模型的思维链推理，并综合各专家裁判的输出，融合了“专家混合”与“自一致性”策略。作为中间件，它在规划模块生成的方案抵达MCP服务器（进而到达机器人底层控制）之前进行门控：方案被**批准**、**驳回重拟**，或**升级至人工审查**。

主要发现与创新点

实验结果显示，该系统在“接受/升级/拒绝”三类决策上达到近**85%的精确率**，对对抗性攻击的**遏制率达97%**，且接受与拒绝任务之间的误判极小，误差主要出现在升级边界处。

核心贡献

本研究首次将LLM裁判集成作为机器人规划的**安全验证中间层**，填补了规划与执行之间安全校验的空白，为机器人自主系统的安全部署提供了可扩展的防护机制，兼具对抗鲁棒性与伦理合规性。

English Summary

Core Contribution

This paper introduces a **novel LLM-driven verification layer** positioned between robot planning and execution, addressing the critical gap in action feasibility validation. Unlike prior work focused solely on task execution, this middleware evaluates action permissibility before plans reach low-level controls.

Method

The system employs an **LLM-as-a-Judge ensemble** that combines chain-of-thought reasoning across multiple models, integrating mixture-of-experts and self-consistency strategies. It gates plans through three outcomes: **approve**, **reject for reformulation**, or **escalate to human review**.

Key Results

Experiments demonstrate **~85% precision** across all decision categories, **97% containment of adversarial attacks**, with minimal errors between accept/reject decisions—errors mainly concentrate at the escalation boundary.

Significance

This work pioneers a scalable safety mechanism for robot autonomy, enhancing both adversarial robustness and ethical compliance while maintaining operational efficiency. The framework offers a practical pathway toward trustworthy autonomous systems in real-world deployments.

Paper 11

Multi-Agent AI Safety as an Institutional Design Problem

Authors: Abdullah X

Categories: cs.LG, cs.AI, cs.MA

URL: <https://arxiv.org/abs/2608.09828v1>

Keywords: system, multi-agent

Source

Abstract

AI agents increasingly work inside systems that govern how they delegate tasks, move information, execute actions, and use shared resources. Recent work already shows that deployment rules can change collective behavior. Here we ask which parts of an AI institution produce safety and how they do it. This is the first paper from POLIS, an ongoing research programme studying algorithmic institutions for multi-agent systems. We report a frozen 5,280-episode study suite. The main pre-specified delegation experiment spans four model families; a targeted high-conflict diagnostic adds three additional model endpoints. In matched structured workflows, the model sees different rule formulations and guards consult different authority states. We also vary the attractiveness of the immediate compliant internal/self fallback and allow blocked workflows to continue. A detailed constitutional prompt produces 0/384 realized violations. A provenance-aware executable guard also produces 0/384, although it blocks prohibited attempts in 51/384 episodes; 44/51 of those episodes later complete safely. The local-state guard's failures concentrate in scenarios where an ordinary transformation changes visible policy while originating authority stays fixed. In matched laundering scenarios, that guard admits violations in 22/96 episodes and provenance enforcement in 0/96 ($p = 4.77 \times 10^{-7}$). A separate resource-allocation experiment shows that revealing the numerical value of an otherwise identical cap changes agent requests. In these structured workflows, the same final violation rate can hide very different mechanisms. The rule itself is only part of the institution. The authority state the system trusts matters, and so does the path available after a block.

Chinese Summary

研究背景与问题

AI智能体正日益运行于复杂的制度性系统中，这些系统规范其任务委派、信息流转、行动执行及资源共享方式。已有研究表明部署规则能改变集体行为，但**制度中哪些具体要素保障安全、如何发挥作用**仍属未知。本文是POLIS研究项目的首篇论文，系统研究多智能体系统的算法制度设计。

方法与实验设计

我们构建了包含**5,280个固定情景**的测试套件，核心委派实验覆盖4个模型家族，另设高冲突诊断场景增加3个模型端点。实验设计关键变量包括：规则表述方式差异、守卫所依据的权威状态、合规内部回退方案的吸引力，以及允许被阻止流程继续运行的条件。同时开展资源分配实验，测试揭示数值上限对智能体请求行为的影响。

主要发现

-
 详细宪法式提示词实现0/384违规，**溯源感知可执行守卫**同样实现0/384违规，但后者在51/384情景中阻止了违规尝试，其中44/51后续安全完成。

-
 局部状态守卫的失败集中于：普通转换改变可见策略但原始权威不变的情景。在匹配的洗钱场景中，该守卫在22/96情景放行违规，而溯源强制为0/96 ($p=4.77 \times 10^{-7}$)。

- 资源分配实验显示，揭示相同上限的数值会改变智能体的请求行为。

核心结论

****相同的最终违规率可能掩盖截然不同的机制**。**规则本身仅是制度的一部分，系统信任的权威状态及被阻止后的可用路径同样关键。制度设计需超越单纯规则优化，关注权威溯源与失败恢复机制。

English Summary

This paper introduces POLIS, a research program on algorithmic institutions for multi-agent systems, investigating which institutional components ensure safety and how. Using a frozen 5,280-episode suite across four model families, we test delegation workflows with varied rule formulations, authority states, and post-block pathways. Key results: a detailed constitutional prompt and a provenance-aware executable guard both achieve 0/384 violations, though the guard blocks 51 prohibited attempts (44 later complete safely). Local-state guards fail specifically when routine transformations alter visible policy while origin authority remains fixed—admitting violations in 22/96 laundering scenarios versus 0/96 with provenance enforcement ($p=4.77 \times 10^{-6}$). A resource-allocation test shows revealing numerical caps changes agent requests. We conclude that identical violation rates can mask fundamentally different mechanisms: institutional safety depends not only on rules but on trusted authority states and available recovery paths after blocking.

Paper 12

Distill Skills into Weights, Not Prompts: Abstract Skills as Privileged Signals for On-Policy Self-Distillation

Authors: Yubo Jiang, Fengying Xie, Zhiguo Jiang, Haopeng Zhang

Categories: cs.LG, cs.AI

URL: <https://arxiv.org/abs/2608.09826v1>

Keywords: reinforcement, learning

Source

Abstract

Reinforcement learning with verifiable rewards yields no group-relative signal when rollout groups are uniformly correct or uniformly wrong, which account for 63.0-68.0% of groups in our experiments. We propose SKALD (Skill-Anchored Latent Distillation), an on-policy self-distillation framework that uses two context views of the same Qwen3-Base model: a question-only student and a teacher conditioned on an abstract, explicit-answer-filtered skill card. The student is trained on its own prefixes, transferring the skill-induced advantage into shared parameters without privileged input at test time. To stabilize context-induced distribution mismatch, SKALD employs an annealed exponentially tilted objective that downweights teacher-preferred tokens with very low student likelihood; as the tilt vanishes, it converges to teacher cross-entropy and recovers the forward-KL student gradient. An empirical gate activates distillation only when verified rollouts estimate a positive teacher advantage. Across five held-out mathematics benchmarks, SKALD improves overall avg@8 over GRPO by +2.46, +4.85, and +12.01 at 0.6B, 1.7B, and 4B, respectively. At 1.7B, zero-variance-only distillation recovers 84.7% of the full gain, while SKALD remains +4.06 above FLOP-matched GRPO and exceeds contextual skill exposure by +3.77. These results show that abstract skills provide dense supervision where group-relative rewards become uninformative.

Chinese Summary

****背景与问题**：**在可验证奖励的强化学习（RL）中，当采样组内所有rollout全部正确或全部错误时，组间相对奖励信号完全失效。实验表明，此类“零方差组”占全部组的63.0%–68.0%，导致模型在这些组上无法获得有效学习信号。

****方法****: 本文提出****SKALD**** (Skill-Anchored Latent Distillation), 一种基于策略的自蒸馏框架。它利用同一Qwen3-Base模型的两个上下文视图: 仅含问题的学生模型, 以及以抽象技能卡 (经显式答案过滤) 为条件输入的教师模型。学生模型仅在自己的前缀上进行训练, 将技能带来的优势迁移到共享参数中, 测试时无需特权输入。为缓解上下文分布不匹配, SKALD采用退火指数倾斜目标函数, 抑制学生似然极低的教师偏好token; 当倾斜系数趋近于零时, 目标退化为教师交叉熵, 恢复前向KL学生梯度。此外, 经验门控机制仅在验证rollout估计出正教师优势时才激活蒸馏。

****主要发现****:

- 在五个保留数学基准上, SKALD相较GRPO在avg@8上分别提升****+2.46 (0.6B) 、+4.85 (1.7B) 、+12.01 (4B) ****。

- 在1.7B规模下, 仅使用零方差组进行蒸馏即可恢复完整增益的****84.7%****; SKALD仍比FLOP匹配的GRPO高****+4.06****, 比上下文技能暴露法高****+3.77****。

- 结果表明, 抽象技能能在组间相对奖励失效时提供密集监督信号, 显著提升学习效率。

****创新点****: 首次将抽象技能作为特权信号用于策略自蒸馏, 并通过退火倾斜与门控机制稳定训练, 实现免特权输入的技能内化。

English Summary

Reinforcement learning with verifiable rewards loses group-relative signals when rollout groups are uniformly correct or wrong—accounting for 63.0–68.0% of groups in our experiments. We propose ****SKALD**** (Skill-Anchored Latent Distillation), an on-policy self-distillation framework using two context views of the same Qwen3-Base model: a question-only student and a teacher conditioned on an abstract, answer-filtered skill card. The student trains on its own prefixes, transferring skill-induced advantages into shared weights without privileged test-time inputs. An annealed exponentially tilted objective stabilizes context mismatch, converging to teacher cross-entropy and recovering forward-KL student gradients; an empirical gate activates distillation only when verified rollouts estimate positive teacher advantage. Across five held-out math benchmarks, SKALD improves avg@8 over GRPO by +2.46, +4.85, and +12.01 at 0.6B, 1.7B, and 4B scales. At 1.7B, zero-variance-only distillation recovers 84.7% of the full gain, while SKALD remains +4.06 above FLOP-matched GRPO and +3.77 above contextual skill exposure. Abstract skills thus provide dense supervision exactly where group-relative rewards become uninformative.

Paper 13

MedPixel: A Unified Pixel-Language Model for Medical Reasoning and Segmentation

Authors: Haoyu Yang, Meixing Shi, Zengjie Chen, Haoran Sun, Haitao Leng et al.

Categories: cs.CV, cs.AI

URL: <https://arxiv.org/abs/2608.09818v1>

Keywords: segmentation, code, image, generation

Abstract

Reliable medical image understanding requires models to connect clinical language and visual reasoning with pixel-level grounding. Yet medical vision-language models often lack precise localization, whereas medical segmenters typically rely on explicit target categories or precise spatial prompts. This divide is reinforced by a supervision mismatch: segmentation datasets provide precise masks but little language supervision, whereas medical vision-language data rarely pair language with dense spatial annotations. To address this gap, we present MedPixel, a unified medical pixel-language model built around a shared language--mask interface. To provide scalable supervision, we introduce MedPLG-440K, comprising approximately 440K pixel-language task samples constructed through a clinically motivated synthesis process without external LLM annotation.

MedPixel is trained with joint multi-task supervised fine-tuning followed by Pixel-Level Preference Optimization, which uses ground-truth masks as offline verifiers to derive response preferences from mask quality. MedPixel supports a broad spectrum of tasks spanning explicit grounding, implicit reasoning, spatial interaction, grounded explanation, and medical VQA. Across this task spectrum, MedPixel achieves strong performance in both pixel-level prediction and response generation, together with effective zero-shot transfer to external grounding benchmarks and robustness to imperfect spatial prompts. Code and model checkpoints will be released at <https://github.com/yhy-whu/Medpixel>.

Chinese Summary

研究背景与挑战

医学图像理解要求模型能够将临床语言、视觉推理与像素级定位相结合。然而，现有医学视觉-语言模型缺乏精准定位能力，而医学图像分割模型又通常依赖显式目标类别或精确的空间提示。这一割裂源于监督信号的不匹配：分割数据集提供精确掩码但缺乏语言监督，而医学视觉-语言数据则很少将语言与密集空间标注配对。

方法创新

本文提出

MedPixel，一种统一的医学像素-语言模型，其核心是构建**共享的语言-掩码接口**，实现语言理解与像素预测的深度融合。为提供可扩展的监督信号，作者引入 **MedPLG-440K** 数据集，包含约44万像素-语言任务样本，通过临床动机驱动的合成流程构建，无需外部大语言模型标注。

训练策略包括两个阶段：

1. **联合多任务监督微调**：覆盖显式定位、隐式推理、空间交互、可解释分割及医学VQA等多种任务
2. **像素级偏好优化**：利用真实掩码作为离线验证器，根据掩码质量推导响应偏好

主要成果

MedPixel 在像素级预测和响应生成方面均取得优异性能，并展现出：

- **零样本迁移能力**：可有效泛化至外部定位基准
- **鲁棒性**：对不完美的空间提示具有较强抗干扰能力

代码与模型将开源至 <https://github.com/yhy-whu/Medpixel>

English Summary

Core Contribution

We present **MedPixel**, a unified medical pixel-language model that bridges clinical language understanding and pixel-level segmentation through a shared language-mask interface. To overcome the supervision mismatch between segmentation and vision-language datasets, we introduce **MedPLG-440K**, approximately 440K pixel-language task samples generated via a clinically motivated synthesis pipeline without external LLM annotation.

Key Innovation

MedPixel employs joint multi-task supervised fine-tuning followed by **Pixel-Level Preference Optimization**, which leverages ground-truth masks as offline verifiers to derive response preferences from mask quality. This enables the model to handle a broad task spectrum, including explicit grounding, implicit reasoning, spatial interaction, grounded explanation, and medical VQA.

Results

MedPixel achieves strong performance across both pixel-level prediction and response generation, demonstrating effective zero-shot transfer to external grounding benchmarks and robustness to imperfect spatial prompts. Code and checkpoints will

be released.

Paper 14

Parameter Exploration for RLVR via Variational Learning

Authors: Vatsal Venkatkrishna, Nico Daheim, Iryna Gurevych

Categories: cs.LG, cs.AI, cs.CL

URL: <https://arxiv.org/abs/2608.09805v1>

Keywords: reinforcement, learning, code, generation

Abstract

Exploration has been a focus of reinforcement learning research for a long time. Recently, there has been growing evidence that it is also an important ingredient in LLM reinforcement learning recipes that can significantly impact downstream performance. Many existing methods control exploration in the action-space, for example, using temperature scaling. However, these methods cannot reorder tokens but only influence the variance in the output distribution. This limits exploration and can lead to divergence or stalled training. Here, we investigate parameter-space exploration, where rollouts are generated by sampling different policies from a posterior that may each explore different rollouts. Sampling less or more diverse policies is then a complementary control lever over exploration. We introduce a family of methods called Perturbed Parameter Policy Optimization (3PO) which use different sampling strategies and different rollout grouping for reward estimation. Experiments on OLMo-3-1025-7B and Qwen2.5-Math-7B across mathematical reasoning and code generation tasks show that these approaches consistently improve average downstream performance over standard GRPO at a near-identical FLOPs cost. Moreover, using multiple parameter samples consistently produces fewer zero-advantage groups and malformed or incorrect rollouts during training than GRPO and action-space baselines. Overall, our work presents evidence that parameter-space exploration can improve reinforcement learning for LLMs.

Chinese Summary

研究背景与问题

强化学习（RL）在大型语言模型（LLM）训练中日益重要，而**探索（exploration）**

是影响下游性能的关键因素。现有方法多在**动作空间**（action-space）中控制探索，如通过温度缩放调节输出分布的方差。然而，这类方法**无法重新排序token**，只能改变分布波动幅度，导致探索受限，易引发训练发散或停滞。

方法创新

本文提出**参数空间探索（parameter-space exploration）**

策略：通过后验分布中采样不同的策略参数来生成rollouts，从而让不同策略探索不同轨迹。通过调整采样策略的多样性，可作为动作空间探索的**互补控制杠杆**。作者提出了一族方法——**扰动参数策略优化（3PO, Perturbed Parameter Policy Optimization）**，包含多种采样策略和不同的rollout分组方式用于奖励估计。

主要发现

在**OLMo-3-1025-7B**和**Qwen2.5-Math-7B**模型上，针对**数学推理**和**代码生成**任务进行实验，结果显示：

- 3PO方法在**近乎相同的FLOPs成本**下，平均下游性能**一致优于**标准GRPO基线；

-

与GRPO和动作空间基线相比，使用多个参数采样**显著减少**了零优势（zero-advantage）分组以及训练中畸形或错误rollouts的出现频率。

结论与意义

本工作提供了**参数空间探索可有效改进LLM强化学习**的实证证据，为RLVR（Reinforcement Learning with Verifiable Rewards）提供了一种新的、更稳健的探索维度，有望成为现有动作空间方法的重要补充。

English Summary

Core Contribution

This paper introduces **parameter-space exploration** for LLM reinforcement learning, complementing traditional action-space methods (e.g., temperature scaling) that cannot reorder tokens and thus limit exploration diversity. The authors propose **3PO (Perturbed Parameter Policy Optimization)**, a family of methods that sample diverse policy parameters from a posterior to generate rollouts, with different sampling strategies and rollout grouping for reward estimation.

Key Results

Experiments on OLMo-3-1025-7B and Qwen2.5-Math-7B across mathematical reasoning and code generation show that 3PO **consistently outperforms standard GRPO** in average downstream performance at nearly identical FLOPs cost. Moreover, multi-parameter sampling **substantially reduces** zero-advantage groups and malformed rollouts during training compared to GRPO and action-space baselines.

Significance

This work provides strong evidence that parameter-space exploration is a viable and effective lever for improving RLVR, offering a more robust exploration mechanism that could mitigate training instability in LLM reinforcement learning.

Paper 15

KGCaRe: Explainable Complex Conditional Question Answering using Automatic Knowledge Graph Construction and Context Retrieval with LLMs

Authors: Ghanshyam Verma, Simanta Sarkar, Devishree Pillai, Hotaka Shiokawa, Yourong Xu et al.

Categories: cs.CL, cs.AI

URL: <https://arxiv.org/abs/2608.09779v1>

Keywords: text, retrieval, knowledge, code, generation, graph, information

Abstract

Answering complex conditional questions using Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) remains a challenge, particularly in domain-specific contexts where general-purpose LLMs and RAG tend to underperform. We hypothesize that augmenting RAG with unstructured and structured knowledge, extracted from both documents and knowledge graphs (KGs), can improve reasoning and answer accuracy for such tasks. To test this, we propose KGCaRe, a hybrid approach that combines neural retrieval with symbolic reasoning over LLM-generated KGs. KGCaRe constructs a KG from documents using a multi-prompt extraction strategy and stores it in a graph database. Simultaneously, the documents are embedded into a vector store to enable neural retrieval. KGCaRe performs innovative iterative graph traversal guided by the LLM to extract relevant triples, prune irrelevant information, and uses additional clue entities to traverse the graph again if the initial traversal does not provide satisfactory context to generate the answer. The relevant triples extracted from the KG in path form, along with semantically retrieved text passages, are then fed into custom KGCaRe prompts to generate answers to the complex conditional questions with explanations. We evaluate KGCaRe on two complex conditional QA datasets. Our

results on these datasets show that KGCaRe consistently outperforms existing baselines, including Vanilla LLM, Code Prompt, Text Prompt, Think-on-Graph, Vanilla RAG, and HybridContextQA, across multiple LLMs such as Mistral, Mixtral, GPT-3.5, and GPT-4o. We publicly release the software pipeline that we developed to implement the proposed KGCaRe approach.

Chinese Summary

研究背景与挑战

利用大型语言模型（LLMs）和检索增强生成（RAG）回答复杂条件性问题仍具挑战，尤其在领域特定场景下，通用模型表现欠佳。现有方法难以同时利用文档中的非结构化知识与知识图谱（KG）中的结构化关系进行联合推理。

方法创新

本文提出 **KGCaRe** 混合框架，核心创新包括：

- **自动知识图谱构建**：通过多提示（multi-prompt）策略从文档中抽取实体与关系，构建领域KG并存入图数据库
- **双通道检索**：将文档嵌入向量库实现神经语义检索，同时利用LLM引导的**迭代图遍历**算法，以路径形式提取相关三元组
- **动态线索实体机制**：当首次图遍历未提供足够上下文时，自动识别“线索实体”触发二次遍历，有效剪枝无关信息
- **可解释答案生成**：将图路径与语义文本片段融合进定制提示模板，生成带推理依据的答案

主要结果

在两类复杂条件问答数据集上，KGCaRe在Mistral、Mixtral、GPT-3.5和GPT-4o等多种LLM下，均稳定超越Vanilla LLM、Vanilla RAG、Think-on-Graph等基线方法，验证了“结构化+非结构化知识联合增强”的有效性。该工作已开源完整软件流水线，为复杂问答提供可复现的解决方案。

English Summary

Core Contribution

We propose **KGCaRe**, a hybrid framework that enhances complex conditional question answering by combining neural retrieval with symbolic reasoning over LLM-generated knowledge graphs. The system automatically constructs domain-specific KGs from documents via multi-prompt extraction, stores them in a graph database, and employs LLM-guided iterative graph traversal with a novel "clue entity" mechanism to extract relevant triples while pruning noise. Retrieved graph paths and semantic text passages are jointly fed into custom prompts to generate explainable answers.

Key Findings

Evaluated on two complex conditional QA datasets, KGCaRe consistently outperforms existing baselines (including Vanilla LLM, RAG, and Think-on-Graph) across four LLMs (Mistral, Mixtral, GPT-3.5, GPT-4o). The results demonstrate that augmenting RAG with both unstructured and structured knowledge significantly improves reasoning accuracy. The complete software pipeline is publicly released to facilitate reproducibility and further research.

Paper 16

MoNo: Multiscale Optimal Transport Neural Operator for Solving PDEs on General Geometries

Authors: Zijiang Yang, Xiaomeng Wu, Dongmei Fu

Categories: cs.LG, cs.AI

URL: <https://arxiv.org/abs/2608.09764v1>

Keywords: transformer

Source

Abstract

Transformer-based neural operators have achieved substantial progress in solving Partial Differential Equations (PDEs) by projecting spatial observations into compact latent tokens and learning physical interactions in latent spaces. However, we reveal that existing learnable projection mechanisms cannot ensure stable and balanced assignments from observation points to latent tokens, causing some latent tokens to be over-assigned while others remain underutilized. This limitation further restricts the design of hierarchical architectures, as assignment imbalance is continuously inherited and amplified across latent spaces, eventually causing severe token collapse in deeper spaces. To address these issues, we propose MoNo (Multiscale Optimal Transport Neural Operator), a progressive multiscale neural operator that efficiently solves PDEs on general geometries through stable latent-space construction. At its core is CoTAP (Cross-scale Optimal Transport Assignment and Projection), a novel latent-space construction method that formulates cross-space assignment between adjacent spaces as an entropy-regularized optimal transport problem, thereby constructing balanced bidirectional projections and stable latent spaces. CoTAP also ensures stable information transfer across multiple latent spaces, further enabling multiscale architectures on general geometries, which in turn support more efficient learning of long-range physical interactions. Extensive experiments demonstrate that MoNo outperforms existing state-of-the-art neural operators in both prediction performance and computational efficiency. Code is available at <https://github.com/ZijiangY1116/MoNo>.

Chinese Summary

研究背景与问题

基于Transformer的神经算子通过在潜在空间中学习物理交互，已在求解偏微分方程（PDE）方面取得显著进展。然而，本研究发现有可学习投影机制无法确保观测点到潜在token的稳定、均衡分配，导致部分token过载而其他token利用率不足。这一缺陷限制了层次化架构设计，因为分配不平衡会在跨尺度潜在空间中被持续继承并放大，最终在深层空间中引发严重的token坍塌。

方法创新

针对上述问题，本文提出**MoNo（多尺度最优传输神经算子）**，一种通过稳定潜在空间构建高效求解通用几何上PDE的渐进式多尺度架构。其核心是**CoTAP（跨尺度最优传输分配与投影）**模块，该模块将相邻潜在空间间的跨空间分配建模为熵正则化最优传输问题，从而构建：

- **均衡的双向投影**：确保观测点均匀映射到各潜在token
- **稳定的潜在空间**：避免token坍塌并保持信息完整性
- **跨尺度稳定信息传递**：支持通用几何上的多尺度架构，高效学习长程物理交互

主要结果

大量实验表明，MoNo在预测精度和计算效率上均优于现有最先进神经算子。代码已开源：<https://github.com/ZijiangY1116/MoNo>

English Summary

Core Contribution

We propose **MoNo**, a progressive multiscale neural operator that solves PDEs on general geometries via stable latent-space construction. Its key innovation, **CoTAP**, formulates cross-scale token assignment as entropy-regularized optimal transport, ensuring balanced bidirectional projections and preventing token collapse in deep latent spaces. This enables efficient hierarchical learning of long-range interactions. Experiments show MoNo outperforms state-of-the-art neural operators in both accuracy and computational efficiency. Code: <https://github.com/ZijiangY1116/MoNo>

Paper 17

Defining Decentralization: An Ontological Perspective

Authors: Jakub Kacper Szelg, Aydin Abadi, Mohammad Naseri

Categories: cs.DC, cs.AI, cs.LG, cs.LO, eess.SY

URL: <https://arxiv.org/abs/2608.09748v1>

Keywords: artificial, system, learning, intelligence, machine, cloud, computing, distributed

Source

Abstract

Decentralization as a concept in computer science has existed for over half a century. Despite its fundamental role across domains such as security, distributed computing, artificial intelligence, cloud infrastructures, and Internet of Things (IoT) architectures, there remains no universally accepted definition of decentralization applicable across computer communication systems. This has become increasingly problematic with the emergence of decentralized AI and machine learning paradigms, including collaborative training, distributed inference, blockchain-based, and agentic AI, where decentralization is often treated as a core design objective. Meanwhile, existing approaches frequently conflate decentralization with related notions such as distribution of trust or specific implementation paradigms. Such ambiguity creates inconsistencies in system analysis, limits comparability between works, and weakens the rigor of formal reasoning surrounding communication architectures and protocol design. In this work, we define this research gap as the Decentralization Problem. We analyze the formal-semantic, epistemological, and pragmatic foundations of decentralization and introduce a graph-based ontology defining it as both relational and subject-specific property of computer communication systems. The framework formally distinguishes decentralization from distribution and supports evaluation through two novel metrics: Void Tolerance and Imperviousness. We also provide a browser-based implementation that enables automated classification and metric computation of arbitrary systems. Instantiations to federated learning and blockchain architectures show consistent, comparable assessments where existing definitions produce incomplete or contradictory conclusions, providing a domain-independent foundation for analysing decentralization across heterogeneous systems.

Chinese Summary

去中心化的定义：一种本体论视角

****背景与问题：****

去中心化作为计算机科学的核心概念已存在半个多世纪，在安全、分布式计算、人工智能、云基础设施及物联网架构等领域具有基础性地位。然而，学界至今缺乏一个跨计算机通信系统普遍适用的统一定义。随着去中心化AI与机器学习范式的兴起（如协作训练、分布式推理、基于区块链及智能体AI），去中心化常被视为核心设计目标，但现有研究往往将其与“信任分布”或特定实现范式混为一谈。这种模糊性导致系统分析不一致、工作间可比性受限，并削弱了通信架构与协议设计形式推理的严谨性。本文将这一研究空白定义为****“去中心化问题”****。

****方法与框架：****

我们从形式语义学、认识论和实用主义三个维度剖析去中心化的基础，提出一种****基于图的本体论框架****，将去中心化定义为计算机通信系统的****关系性****与****主体特定性****属性。该框架在形式上区分了去中心化与分布性，并引入两个新度量指标：****空穴容限（Void Tolerance）****与

****不可渗透性（Imperviousness）****。我们同时提供了一个基于浏览器的实现工具，可对任意系统进行自动化分类与度量计算。

****主要发现与创新：****

将该框架应用于联邦学习与区块链架构的实例化评估表明，它能产生一致、可比较的结果，而现有定义在这些场景下往往给出不完整甚

至矛盾的结论。本工作为跨异构系统分析去中心化提供了**领域无关的形式化基础**，填补了该领域长期存在的理论空白。

English Summary

Defining Decentralization: An Ontological Perspective

Motivation: Despite half a century of use in computer science, decentralization lacks a universally accepted definition across communication systems, creating inconsistencies in analysis and weakening formal reasoning—especially critical with the rise of decentralized AI paradigms.

Approach: We formalize this gap as the *Decentralization Problem* and propose a *graph-based ontology* that defines decentralization as a *relational, subject-specific property* of systems, formally distinguishing it from distribution. Two novel metrics—*Void Tolerance* and *Imperviousness*—enable quantitative evaluation, supported by a browser-based tool for automated classification.

Results: Instantiations on federated learning and blockchain demonstrate consistent, comparable assessments where prior definitions fail or conflict. This work provides a *domain-independent foundation* for analyzing decentralization across heterogeneous systems, addressing a critical theoretical gap in computer communication research.

Paper 18

SR-OPSD: Self-Referenced On-Policy Self-Distillation

Authors: Zhuo Sun, Entong Li, Yanlong Zhao, Xiaoyuan Cheng, Wenxuan Yuan et al.

Categories: cs.LG, cs.AI, stat.ML

URL: <https://arxiv.org/abs/2608.09745v1>

Keywords: reinforcement, learning

Source

Abstract

On-policy self-distillation (OPSD) converts feedback into dense token-level supervision on trajectories generated by the policy to be optimized, providing a useful complement to reinforcement learning with sparse outcome rewards. However, the self-teacher policy used in OPSD is typically a stop-gradient or exponential-moving-average copy of the policy conditioned on additional context information, and thus co-evolves with both the student policy and its on-policy context distribution. Directly matching such a moving target with a fixed projection objective can lead to unstable optimization or excessive distributional concentration. This nature of OPSD motivates the proposed *Self-Referenced On-Policy Self-Distillation (SR-OPSD)*. At fixed student-generated contexts, a token-level variational characterization identifies the effective distillation target as a geometric interpolation between the self-teacher policy and a reference policy. Meanwhile, we use the Rényi divergence family to generalize the projection geometry. This formulation separates *where* the adaptive target is placed from *how* the student is projected toward it: the interpolation coefficient controls underlying target, while the Rényi order controls the projection geometry and its sensitivity to token-level density ratios. Extensive experiments across scientific evaluation, mathematical reasoning, and coding generation tasks with multiple large language models show that SR-OPSD achieves the state-of-the-art or competitive performance across various settings.

Chinese Summary

研究背景

在线策略自蒸馏（OPSD）将反馈转化为密集的token级监督信号，用于优化策略生成的轨迹，是对稀疏奖励强化学习的有效补充。然而，OPSD中的“自教师”策略通常是基于额外上下文信息的学生策略的停止梯度或指数移动平均副本，因此会与学生策略及其上下文分布共同演化。直接使用固定投影目标匹配这种动态目标，可能导致优化不稳定或分布过度集中。

方法创新

针对上述问题，本文提出**自参考在线策略自蒸馏（SR-OPSD）**方法，核心创新包括两点：

1.

变分目标分解：在固定的学生生成上下文下，通过token级变分特征化，将有效蒸馏目标表示为自教师策略与参考策略之间的**几何插值**，从而明确区分“自适应目标的位置”与“学生向目标投影的方式”。

2.

Rényi散度泛化：引入Rényi散度族来泛化投影几何。插值系数控制底层目标，而Rényi阶数控制投影几何及其对token级密度比的敏感性，实现更灵活的优化控制。

实验结果

在科学评估、数学推理和代码生成等多项任务上，基于多种大语言模型的广泛实验表明，SR-OPSD在多种设置下均达到**最先进或具有竞争力的性能**，验证了其有效性和鲁棒性。

English Summary

Core Contribution

This paper proposes **SR-OPSD (Self-Referenced On-Policy Self-Distillation)** to address the instability of standard on-policy self-distillation, where the self-teacher policy co-evolves with the student, making fixed projection objectives unreliable.

Method

SR-OPSD introduces a token-level variational characterization that formulates the effective distillation target as a **geometric interpolation** between the self-teacher and a reference policy. It further generalizes the projection geometry using the **Rényi divergence family**, decoupling *where* the adaptive target lies (controlled by the interpolation coefficient) from *how* the student is projected (controlled by the Rényi order).

Results

Extensive experiments across scientific evaluation, mathematical reasoning, and code generation with multiple LLMs demonstrate that SR-OPSD achieves **state-of-the-art or competitive performance** in various settings, confirming its robustness and effectiveness.

Paper 19

Rethinking Factor Sharing in Federated LoRA: A Rank-Aware Adaptive Approach

Authors: Xinyi Xu, Bingnan Xiao, Shuang Qin, Gang Feng, Tony Q. S. Quek

Categories: cs.LG, cs.AI, cs.DC

URL: <https://arxiv.org/abs/2608.09742v1>

Keywords: model, large, learning, language, federated

Abstract

Low-rank adaptation (LoRA) represents large language model (LLM) updates with two compact matrix factors, i.e., A and B , providing an efficient way to fine-tune large models in federated learning paradigm. Inspired by the asymmetric roles of the LoRA factors, we study whether A should be shared across clients while B remains client-specific (Share-A/Local-B), or whether B should instead be shared while A remains client-specific (Share-B/Local-A). With a least-squares surrogate, we reveal that Share-A/Local-B requires the client-specific LoRA update matrices to use a common rank- r input-side space, whereas Share-B/Local-A requires a common rank- r output-side space. The two strategies therefore incur different projection residuals, indicating that the preferred strategy is the one with the smaller aggregate residual across clients. With this insight, we propose Federated Adaptive Factor Sharing Low-Rank Adaptation (FedAS-LoRA), which selects the sharing side before training to enhance fine-tuning performance. To enable adaptive factor selection before training, we design a Rank-Aware Shared-Subspace Sufficiency (RSS) metric, which effectively assesses whether a shared rank- r input subspace is sufficient for the local data distributions using representations extracted from a frozen LLM backbone. Experiments across different tasks, data distributions, LoRA ranks, and participation settings confirm the effectiveness of RSS and the superior performance of FedAS-LoRA.

Chinese Summary

研究背景与问题

低秩适配 (LoRA) 通过两个紧凑矩阵因子 (A 和 B) 表示大语言模型 (LLM) 的参数更新, 为联邦学习中的高效微调提供了可行方案。然而, 现有联邦LoRA方法通常默认共享某一因子, 缺乏对因子共享策略的理论分析与自适应选择机制。

核心发现

本文基于最小二乘代理模型, 首次从理论上揭示了两种因子共享策略的本质差异:

- **共享A/本地B (Share-A/Local-B)**: 要求各客户端的LoRA更新矩阵共享相同的秩- r 输入侧子空间。
- **共享B/本地A (Share-B/Local-A)**: 则要求共享秩- r 输出侧子空间。

两种策略会产生不同的投影残差, 因此**最优策略应为跨客户端聚合残差更小的那一种**。这一发现推翻了“固定共享某一因子”的直觉, 指出了因子共享方向需根据数据分布自适应决定。

提出的方法

基于上述理论, 本文提出**联邦自适应因子共享低秩适配 (FedAS-LoRA)**, 其核心创新包括:

1. **训练前自适应选择共享侧**: 根据客户端数据分布特征, 在训练开始前即确定共享 A 还是共享 B , 以最小化总体投影残差。
2. **秩感知共享子空间充分性 (RSS) 指标**: 利用冻结的LLM骨干网络提取表示, 无需训练即可评估共享秩- r 输入子空间对本地数据分布的充分性, 从而指导因子选择。

实验验证

在多种任务、数据分布 (IID/Non-IID)、LoRA秩和客户端参与率设置下, 实验结果表明:

- RSS指标能有效预测最优共享策略;
- FedAS-LoRA相比固定共享策略 (如Share-A或Share-B) 取得显著更优的微调性能, 验证了自适应选择的必要性与有效性。

English Summary

Core Contribution

This paper rethinks factor-sharing strategies in federated LoRA by theoretically proving that **Share-A/Local-B** and **Share-B/Local-A** require a common rank- r input-side vs. output-side subspace, respectively, leading to different projection residuals. The optimal strategy is the one minimizing the aggregate residual across clients.

Proposed Method

We introduce **FedAS-LoRA**, which adaptively selects the sharing side *before training* using a novel **Rank-Aware Shared-Subspace Sufficiency (RSS)** metric. RSS leverages frozen LLM representations to assess whether a shared rank- r input subspace suffices for local data distributions, enabling data-driven factor selection without training overhead.

Key Results

Experiments across diverse tasks, data distributions, LoRA ranks, and participation settings confirm that RSS accurately predicts the preferred strategy, and FedAS-LoRA consistently outperforms fixed-sharing baselines, demonstrating the practical value of adaptive factor sharing in federated fine-tuning.

Paper 20

Model Discovery Agent: LLM-assisted Bayesian experiment design for data-efficient discovery of mechanistic world models

Authors: Kevin Murphy

Categories: cs.AI

URL: <https://arxiv.org/abs/2608.09696v1>

Keywords: model, language, large

Source

Abstract

Predicting the answer to interventional “what if” questions --- the outcome of an action never taken --- requires a **mechanistic**, causal model, not a curve fit; and learning such a model requires **experiments**, because passive data leaves its mechanisms unidentified. Experiments are expensive, so the central problem is **data efficiency**. We present the Model Discovery Agent (MDA), which couples a large language model (LLM), used as a **proposer** of candidate structures, with standard Bayesian machinery --- sequential Monte Carlo (SMC) for parameter and structure posteriors, simulation-based inference (SBI) for intractable likelihoods, and value-of-information (VoI) for experiment design --- to discover latent mechanistic world models from few interventions. MDA operates in the M-open setting: when the truth lies outside the current hypothesis class, a predictive check flags the inadequacy and the proposer expands the hypothesis space with a new model whose parameters are then identified by designed experiments. We show that **discovery and design reinforce**: the design step identifies the mechanism the discovery step proposes, and the identified mechanism improves predictions, enabling further discoveries from the remaining unexplained residuals. On three different benchmarks --- covering physics (**DPbench**, [wiemann2026discoverphysics](#)), chemistry (**CHEMbench**, [kabra2026autoscilab](#)) and biology (**HHbench**, a new partially observed single-neuron electrophysiology benchmark we create) --- we show that MDA sets a new SOTA in terms of data-efficient model learning and reliable interventional forecasting ability.

Chinese Summary

背景与问题：预测干预性“假设”问题（即从未执行行动的结果）需要**机制性因果模型**，而非简单的曲线拟合。被动数据无法识别机制，因此必须依赖**实验**，但实验成本高昂，**数据效率**成为核心挑战。

方法创新：本文提出**模型发现智能体（MDA）**，将大语言模型（LLM）作为**候选结构生成器**，与标准贝叶斯工具链深度融合：

- **序贯蒙特卡洛（SMC）** 用于参数与结构后验推断

- **基于模拟的推断 (SBI)** 处理不可解似然
- **信息价值 (Vol)** 指导实验设计

MDA在**M-open**设定下运行：当真实模型超出当前假设类时，预测性检查会发出警告，LLM随即扩展假设空间，提出新模型，随后通过设计的实验识别其参数。

核心发现：**发现与设计相互强化**——设计步骤验证发现步骤提出的机制，而识别出的机制改善预测，进而从剩余残差中促成新发现。

实验验证：在三个基准上（物理的DPbench、化学的CHEMbench、生物学的HHbench——后者为本文新建的部分观测单神经元电生理基准），MDA在数据高效模型学习和可靠干预预测方面均达到**新SOTA**。

关键创新点：

1. LLM作为机制提议者，突破固定假设空间限制
2. 主动实验设计驱动机制识别，而非被动拟合
3. 发现-设计循环实现数据效率的迭代提升

English Summary

Core Contribution: We introduce the **Model Discovery Agent (MDA)**, a framework that couples LLM-based structure proposal with Bayesian machinery (SMC, SBI, and Vol) to discover mechanistic world models from minimal interventions, operating in the M-open setting where the truth may lie outside the current hypothesis class.

Key Mechanism: MDA alternates between predictive checks (flagging model inadequacy), LLM-driven hypothesis expansion, and Vol-optimized experiments. This creates a **discovery-design feedback loop**: designed experiments validate proposed mechanisms, and validated mechanisms improve predictions, enabling further discoveries from residuals.

Results: On three benchmarks (physics DPbench, chemistry CHEMbench, and a new biology benchmark HHbench for partially observed neuron electrophysiology), MDA achieves **state-of-the-art data efficiency** and reliable interventional forecasting, demonstrating that LLM-assisted Bayesian experiment design can substantially reduce experimental cost while maintaining mechanistic fidelity.
